

Wesley DeMontigny

University of Maryland, College Park

wes.demontigny@gmail.com, [GitHub](#), [Portfolio](#)

Research Experience:

Graduate Research Assistant | University of Maryland | 2023-Present

- **Advanced Statistical Modeling:** Led the research and development of complex probabilistic models for heterogeneous and incomplete datasets.
- **Deep Learning:** Implemented GNN, LSTM, normalizing-flow, and metric-learning models in PyTorch for link prediction, time-series forecasting, and simulation-based inference.

Select Projects:

Hybrid Time-Series Forecasting Pipeline ([Write-Up](#))

- Won 1st place in an ASA-sponsored forecasting competition, developing an approach using representation learning and classical statistics to predict peak cherry blossom blooms.
- Pre-trained an LSTM neural network on climate time series and transferred latent representations to bloom-date prediction, improving generalization for a highly data-deficient task.

Probabilistic Imputation Model for Incomplete Genomic Data ([GitHub](#), [Preprint](#))

- Designed and implemented a probabilistic graphical model in Python (JAX/NumPyro) to impute missing values in high-dimensional binary vectors.
- Achieved 2x higher precision than competing methods at comparable recall, and enabled uncertainty quantification for downstream analysis.

Monte Carlo Sampler for Regime-Switching Models ([GitHub](#), Manuscript Forthcoming)

- Engineered a high-performance C++ Reversible-Jump MCMC sampler to detect hidden state transitions in continuous-time stochastic processes and automatically select model complexity.

The Bayesian Battleship Benchmark ([GitHub](#), [Write-Up](#))

- Built an SMC sampler for a partially observed sequential decision problem and compared MAP, Thompson sampling, and information-based Bayesian policies against frontier LLM agents.

Machine Learning Approaches for Gene Content Prediction (Ongoing Work)

- Developed a comparative evaluation study in PyG/PyTorch, contrasting message-passing GNNs (LightGCN, SocialLGN), geometric embedding models (TransR, DistMult), and combined approaches for transductive edge prediction in genome-genome-gene graphs.

Education:

Ph.D. Candidate in Biology, 2023-2027 (Expected), *University of Maryland* GPA: 4.0

- **Focus:** Computational Statistics, Bayesian Inference, and Computational Biology
- **Dissertation Topic:** Imputation Models for High-Dimensional Partially Observed Datasets

B.S. in Biology, 2019-2022, *Georgia College & State University* GPA: 3.95

- **Awards:** Goldwater Scholar (2022) | Commencement Speaker for College of Arts & Sciences

Skills:

- **Languages:** C++, Python, R, SQL
- **Machine Learning & Probabilistic Frameworks:** PyTorch, PyG, JAX, NumPyro, Scikit-Learn, NumPy, SciPy, pandas, Eigen3, OpenMP, Boost
- **Modeling Expertise:** Bayesian Inference, Graphical Modeling, Latent Variable Modeling, Probabilistic Clustering, Deep Learning, Simulation Analyses, Time Series Analysis
- **DevOps & Tools:** Docker (DevContainers), Conda, Git, Linux Shell, CMake